

SOFTWARE • PRODUCT DESIGN REPORT • SIH 2026

E-WASTE BRIDGE

Collector-first digital bridge from informal collection to authorized recycling

DESIGN THESIS

Do not try to replace the informal network. Preserve its strongest advantages—local reach, speed, familiarity and immediate settlement—while adding price transparency, authorized routing and traceability.

REPORT CONTROL	VALUE
Working title	E-Waste Bridge
Document type	Product Design Report (PDR)
Submission context	Smart India Hackathon 2026 • Software
Primary user	Informal e-waste collector / kabadi operator
Secondary user	CPCB-registered / authorized recycler or collection partner
Status	Design baseline + prototype specification
Scope note	Prototype claims are deliberately separated from validation requirements

01 • EXECUTIVE SUMMARY

Executive Summary

India's e-waste challenge is not only a recycling-capacity problem. A critical operational gap exists between the people who already collect discarded electronics at neighborhood level and the formal, traceable recycling ecosystem that is expected to process e-waste responsibly. The E-Waste (Management) Rules, 2022 establish the formal regulatory framework, while CPCB maintains registration and EPR mechanisms for relevant stakeholders. The product opportunity is therefore not another generic disposal app; it is a low-friction transaction layer for the collector who asks: "What is this lot worth, who can take it, how do I hand it over compliantly, and when do I get paid?"

DESIGN OBJECTIVE	SUCCESS CONDITION
Make formal routing as convenient as informal selling	A collector can create, value, route and close a lot with minimal steps and without mandatory digital payment.
Increase price transparency	The collector sees an explainable reference estimate and competing eligible buyers/recyclers.
Increase formal handover	Every completed transaction receives a unique reference and auditable event trail.
Work in real field conditions	Core collection workflows remain usable with intermittent connectivity and entry-level Android devices.
Avoid false automation	AI is advisory/assisted where data is sufficient; rule-based fallback exists for sparse categories.

CORE PROPOSITION

A collector-first marketplace and compliance bridge that connects small e-waste lots to authorized recycling capacity without forcing collectors to abandon cash-first behavior.

02 • PROBLEM DEFINITION

1. Problem Definition and Root Cause

1.1 The current chain

A typical neighborhood e-waste flow can be represented as:

ORIGIN	CURRENT INTERMEDIATION	FORMAL / INFORMAL END POINT
Household / small business	Local collector → aggregator → trader	Authorized recycler (sometimes) or uncontrolled processing (sometimes)
Institution / business	Scrap dealer / aggregator → bulk trader	Recycler / dismantler / refurbisher route

The important design insight is that the informal network is not simply "unorganized." It has evolved around strong economic and operational incentives: immediate settlement, established buyer relationships, local access and low transaction overhead. A digital intervention that removes those advantages is unlikely to achieve adoption.

1.2 Consequence chain

1. Low visibility into formal buyers and compliant handover routes makes the formal channel harder to access.

2. Extra travel, waiting, documentation or uncertainty can make an informal buyer more attractive even where a formal option exists.
3. When e-waste is diverted into unsound processing, workers and surrounding communities can be exposed to hazardous pollutants; WHO identifies open burning, heating, acid leaching and unsafe dismantling among hazardous practices.
4. Uncontrolled processing can also reduce the quality and completeness of material recovery and make downstream traceability difficult.

WHO reports that 62 million tonnes of e-waste were generated globally in 2022, with only 22.3% documented as properly collected and recycled. WHO also describes exposure risks from unsound recycling practices such as open burning and acid leaching. These global findings support the health and environmental relevance of safer routing, while the Indian regulatory context defines the formal actors and registration mechanisms that the product must respect.

ROOT-CAUSE GAP	WHAT IS MISSING	PRODUCT RESPONSE
Informational	Current reference value, eligible buyer/recycler, route options, handover evidence	Photo-assisted lot creation, transparent estimate, ranked eligible destinations
Institutional	A simple collector-facing entry point into the formal chain	Collector account + transaction workflow + recycler interface
Economic	Formal route may have higher perceived friction than immediate informal sale	Cash-first settlement, optional digital payment, local routing/pickup where available
Trust	Existing buyer is familiar; new systems are uncertain	Verified counterparties, visible terms, transaction reference, proof of handover

03 • ECOSYSTEM & REGULATORY CONTEXT

2. Formal-System Context

The E-Waste (Management) Rules, 2022 came into force on 1 April 2023 and superseded the 2016 rules. The rules apply to specified stakeholders involved in manufacture, sale, transfer, purchase, refurbishing, dismantling, recycling and processing of covered electrical and electronic equipment. CPCB guidance states that, under the current regulation, registered Producers, Recyclers and Refurbishers on the EPR portal can collect e-waste. This means the product must be designed as an access and coordination layer into the formal ecosystem—not as a replacement for statutory registration or authorization.

REGULATORY FACT	PDR DESIGN IMPLICATION
CPCB maintains EPR registration / stakeholder mechanisms.	Recycler eligibility must be represented explicitly and validated against the authoritative data source used by the deployment.
CPCB states that registered Producers, Recyclers and Refurbishers can collect under the 2022 rules.	The platform should not represent an unregistered collector as an independently authorized recycler or formal processor.
CPCB guidance requires records and reporting around collection, dismantling, recycling and material movement for relevant registered actors.	The platform should generate structured transaction evidence and preserve an auditable chain of events.
The 2024 amendment introduced provisions for EPR certificate exchange/trading platforms and strengthened the digital compliance ecosystem.	The proposed product should remain interoperable with—not duplicate—official compliance functions.

COMPLIANCE BOUNDARY

The platform is a coordination, transparency and traceability layer. It does not confer legal authorization, replace CPCB registration, or certify a recycler independently of the authoritative registry.

04 • EXISTING LANDSCAPE

3. Existing Landscape and Unsolved Gap

The relevant ecosystem is already populated. The opportunity is created by the way existing services partition the workflow rather than by claiming that no e-waste infrastructure exists.

PLAYER / LAYER	PRIMARY JOB	OBSERVED LIMITATION FOR THIS USE CASE
CPCB EPR ecosystem	Registration, compliance and formal EPR mechanisms	Not designed as a daily, low-literacy transaction workflow for a small informal collector.
PRO / ecosystem operators	Producer compliance, collection programmes and formal reverse logistics	Operate at ecosystem/programme level; the smallest collector's immediate lot-to-buyer decision is not the sole product objective.
B2B marketplaces	Price discovery, sourcing, transactions and documentation	Often optimized for business transactions, larger lots and conventional digital workflows.
Consumer pickup apps	Doorstep disposal / collection convenience	Primary interaction begins with the waste generator, not the collector attempting to route a collected lot downstream.
NGOs / worker formalization programmes	Training, inclusion, capacity building and worker support	Human-intensive and locally dependent; digital transaction infrastructure can complement but not replace these interventions.

Public product information confirms that Karo Sambhav operates collection centres, technology-enabled traceability and EPR-related services, while Recykal positions its marketplace around businesses, recyclers, brands and enterprises with traceability and compliance-ready transactions. Their existence reinforces rather than invalidates the proposed gap: the PDR targets the collector's moment-of-sale workflow and the friction of entering a formal route.

05 • PRODUCT STRATEGY

4. Product Thesis: Formalize by Reducing Friction

NORTH-STAR PRINCIPLE

The informal network should be bridged, not digitally “defeated.” Adoption depends on making the compliant route at least competitive on the dimensions the collector actually values.

COLLECTOR ADVANTAGE TODAY	DIGITAL PRODUCT EQUIVALENT	TRUST MECHANISM
Immediate cash	Cash-first settlement; digital payment optional	Visible settlement status + receipt

COLLECTOR ADVANTAGE TODAY	DIGITAL PRODUCT EQUIVALENT	TRUST MECHANISM
Nearby buyer	Distance-aware destination ranking	Verified recycler identity
Fast decision	Photo/lot capture in a few taps	Explainable estimate + confidence band
Known relationship	Repeat buyer / recycler history	Ratings, completion history, repeat routing
Low paperwork	Generated lot reference and handover record	Offline event log + later synchronization

06 • USERS AND WORKFLOWS

5. Users, Jobs-to-be-Done and End-to-End Workflow

5.1 Primary personas

USER	PRIMARY JOB	DESIGN NEED
Collector / kabadi operator	Convert a mixed or categorized e-waste lot into cash with minimal delay.	Low literacy, local language, weak connectivity, minimal data entry, trust cues.
Recycler / authorized processor	Acquire suitable feedstock with reliable quantity, category, location and traceability.	Structured lot data, eligibility filters, intake capacity, pickup / acceptance workflow.
Platform operator / compliance team	Maintain data quality, verify counterparties, manage disputes and monitor transactions.	Audit trail, exception queue, registry sync, fraud/anomaly signals.

5.2 Collector workflow

STEP	COLLECTOR ACTION	SYSTEM OUTPUT
1 • Capture	Photograph / select item(s) and optionally enter approximate weight.	Lot draft created offline if needed.
2 • Classify	Confirm suggested category; correct visually if needed.	Normalized material/category label + confidence.
3 • Value	Request estimate.	Reference range, not a guaranteed price.
4 • Match	See eligible recyclers / collection partners ranked by rate, distance and pickup availability.	Shortlist with clear trade-offs.
5 • Accept	Select buyer/recycler and terms.	Provisional order / acceptance record.
6 • Handover	Deliver / schedule pickup and confirm quantity.	Timestamp, location (with privacy limits), counterparty, lot reference.
7 • Settle	Receive cash or optional digital payment.	Settlement status + receipt.
8 • Learn	View earnings history and basic safety reminders.	Ledger + repeat workflow.

07 • FUNCTIONAL ARCHITECTURE

6. System Architecture

The architecture is modular so that the collector experience stays simple while the backend manages verification, pricing, transaction state and data quality.

LAYER	COMPONENTS	RESPONSIBILITY
Collector app	Android UI, camera, local database, voice/pictorial guidance, sync engine	Create lots, estimate, compare routes, record handover and settlement offline-first.
Recycler portal	Web app / responsive interface, inventory feed, acceptance queue	Publish buying preferences, accept lots, schedule pickup, confirm weight and settlement.
Core API	Auth, lot service, pricing service, matching service, transaction service	State transitions, permissions, business rules and API contracts.
Verification	Recycler registry connector / periodic authority-source sync, counterparty checks	Keep eligibility metadata current; surface stale or unverifiable records.
Analytics / AI	Classification, price estimation, anomaly scoring	Advisory intelligence; model versioning and confidence.
Data layer	Operational DB + event log + analytical store	Maintain transactional state, history, traceability and training data lineage.

OFFLINE-FIRST RULE

Lot creation, photo capture, category confirmation and local transaction drafting must function without live network access. Cloud sync should be eventual and conflict-aware rather than a prerequisite for every interaction.

SYSTEM FLOW

COLLECTOR DEVICE	PLATFORM SERVICES	RECYCLER / FORMAL END POINT
Photo / lot → local store → estimate → ranked options	Sync → validate → match → transaction state → audit log	Accept → pickup / intake → weigh → process → close
Receipt + ledger visible offline	Event stream stores every state change	Settlement + handover confirmation

08 • DATA ARCHITECTURE

7. Six Mandatory Datasets + AI Training Dataset

The data layer is a core product component rather than a static database. Each dataset needs an explicit source, update mechanism, validation rule and known limitation.

DATASET	MINIMUM FIELDS	GENERATION / UPDATE	QUALITY CONTROL
1. Material Dataset	Category, subcategory, item type, visual cues, hazardous flags, unit conventions	Seed taxonomy; curator updates; verified recycler feedback	Versioned taxonomy; unknown category allowed; no forced classification
2. Price Dataset	Material/category, region, timestamp, quoted rate, accepted rate, unit, source	Recycler quotes + completed transactions + controlled external benchmarks	Outlier detection; freshness score; geography and condition normalization

DATASET	MINIMUM FIELDS	GENERATION / UPDATE	QUALITY CONTROL
3. Recycler Dataset	Entity ID, registration/eligibility status, location, categories accepted, capacity, pickup terms, last verification	Authority-source sync + controlled partner onboarding	Status timestamp; verification source; stale-record flags
4. Transaction Dataset	Lot ID, seller, buyer, category, weight, quote, final price, timestamps, status	Every platform transaction emits structured events	Immutable event references; reconciliation between acceptance and closure
5. Traceability Dataset	Lot genealogy, handover events, location granularity, proof references	Generated automatically at workflow milestones	Event sequence validation; privacy minimization; access controls
6. Collector Dataset	Minimal profile, language, service area, consent/preferences, optional repeat-buyer history	Explicit user entry; progressive profiling	Collect only necessary data; consent logging; deletion/export path where applicable
7. AI/ML Dataset	Image/category labels, price features, outcomes, model feedback, quality labels	Curated training set + validated production feedback	Source/label lineage, train-test separation, versioned models, bias/error monitoring

7.1 Data principles

- Unknown is a valid value. The UI should allow “not sure” and route uncertain lots to assisted verification rather than forcing an incorrect AI label.
- Every model output carries a version and confidence / reliability indicator.
- Price data is time- and geography-sensitive; estimates should display a validity context rather than a single timeless number.
- Personal data is minimized. The product should not collect identity information that is not necessary for transaction, payment, safety or compliance requirements.
- Corrections become data-quality events. A recycler or operator correction should feed back into curation and model evaluation workflows.

09 • AI / ML

8. AI/ML Design — Useful, Bounded and Explainable

USE CASE	MVP APPROACH	FALLBACK	SUCCESS MEASURE
Material classification	Image classifier for common categories where training data is sufficient.	Rule-based cues + manual category selection / assisted review.	Top-1 / top-3 accuracy on held-out field-like images; abstention quality.
Price estimation	Regression / ranking model using category, weight, location, recency and observed quotes.	Reference-rate tables + deterministic adjustments.	Median absolute error / interval coverage; avoid false precision.
Recycler ranking	Rules + scoring: eligibility, distance, rate, capacity, pickup capability, reliability.	Eligibility + distance only.	Acceptance rate, completion rate, average travel burden, price competitiveness.

USE CASE	MVP APPROACH	FALLBACK	SUCCESS MEASURE
Transaction anomaly detection	Rules + lightweight anomaly model over repeated patterns.	Hard thresholds and operator review.	Precision of alerts and reduction in suspicious/duplicate events.

WHAT THE MVP MUST NOT CLAIM

The system does not “guarantee market price,” “identify every material correctly,” or “automatically prove legal authorization.” It provides assisted estimation and routing based on available, time-stamped data.

10 • UX AND ACCESSIBILITY

9. Collector-First UX

DESIGN REQUIREMENT	IMPLEMENTATION DECISION	RATIONALE
Low literacy	Picture-led cards, large tap targets, minimal text, voice prompts and spoken confirmation	Reduces dependency on reading and dense forms.
Vernacular	Hindi + Marathi at minimum for MVP; architecture supports additional locales	Language should be part of interaction design, not a translation layer added later.
Intermittent connectivity	Local queue, offline draft, background sync, visible sync state	Prevents network availability from becoming a hard blocker.
Entry-level Android	Small app footprint, compressed images, lazy loading, limited animations	Keeps target hardware assumptions modest.
Trust	Clear buyer identity, amount/range, terms, settlement status and receipt	Makes the formal route understandable before commitment.
Error recovery	Undo/edit category, duplicate detection, retry sync, assisted escalation	Field workflows are noisy; recovery is more valuable than perfect data entry.

11 • SAFETY

10. Safety as a Workflow Mechanism

Safety guidance is embedded at the moment of handling, not isolated on an informational page. The system can attach pictorial/audio cues to categories that commonly present higher handling risk, while avoiding instruction that encourages hazardous extraction or processing.

TRIGGER	SAFE GUIDANCE PATTERN	PRODUCT BEHAVIOR
Battery-containing lot	Pictorial “do not puncture / heat / open” reminder; escalation path for damaged items.	Require acknowledgement before closing the lot; allow “damaged battery” flag.
CRT / fragile display	Pictorial handling warning; avoid glass breakage.	Safety card appears before handover confirmation.
Loose PCB / mixed electronic parts	Avoid unsound heating / chemical processing; route to eligible recycler.	Offer formal destinations and record handover rather than extraction workflow.

TRIGGER	SAFE GUIDANCE PATTERN	PRODUCT BEHAVIOR
Unknown / hazardous appearance	Do not classify by guess when confidence is low.	Escalate to assisted review / qualified receiving party.

WHO documents hazardous exposure from informal e-waste activities including open burning, heating, acid leaching and unsafe dismantling. The product therefore uses safety prompts to steer material toward compliant downstream processing rather than teaching recovery techniques.

12 • RECYCLER EXPERIENCE

11. Recycler-Side Interface

The recycler portal is a first-class product surface because the collector workflow only works when downstream demand is visible and credible.

SCREEN / MODULE	KEY ELEMENTS	OUTCOME
Demand / intake dashboard	New lots, distance, category, quantity, risk flags, required action	Recycler sees actionable supply instead of a raw marketplace feed.
Buying preferences	Accepted categories, minimum quantity, target geographies, indicative rates, pickup capability	Matching engine can rank relevant destinations.
Lot detail	Photos, classification, estimated weight, offered terms, evidence and event history	Recycler can accept, counter, request clarification or reject with reason.
Pickup / handover	Schedule, driver/agent assignment, proof of pickup, actual weight	Creates a verifiable chain from collection to intake.
Settlement	Final price, deductions if any, settlement mode, receipt	Closes transaction transparently.
Compliance / audit view	Registration metadata, transaction exports, event history, verification timestamps	Supports governance without pretending to replace statutory portals.

13 • SECURITY & PRIVACY

12. Privacy, Security and Data Governance

AREA	DESIGN CONTROL
Identity	Minimal profile; role-based access; do not expose collector personal details unnecessarily.
Location	Use coarse location for matching where precise coordinates are not required; retain exact location only for a justified transaction purpose.
Photos	Compress on device; associate with lot ID; define retention policy; prevent unintended gallery disclosure.
Payments	Store settlement status/reference rather than sensitive payment credentials wherever possible; use established payment providers for digital settlement.
Auditability	Append-only event references for transaction state changes; operator actions logged.

AREA	DESIGN CONTROL
Model governance	Model versions, training-data lineage, confidence thresholds and rollback path.
Offline security	Encrypted local storage for queued transaction data; device logout / session expiry; safe sync retry.

14 • ECONOMICS

13. Unit Economics and Platform Sustainability

The economic objective is to improve the collector's effective outcome after considering price, travel, time, waiting, and payment certainty. Exact rates cannot be responsibly specified until pilot data exists.

VARIABLE	INFORMAL ROUTE	PLATFORM ROUTE TARGET
Headline sale value	Known buyer quote	Competitive reference + eligible offers
Travel cost / time	Often low because buyer is nearby	Minimize through proximity ranking / pickup options
Settlement time	Typically immediate / familiar	Immediate cash-first where operationally available; digital optional
Documentation burden	Low / informal	Automated lot reference + handover record
Price transparency	Low / relationship-based	Reference range + source freshness + buyer comparison
Platform cost to collector	None or embedded in spread	Target: no mandatory upfront fee for collector in MVP
Revenue to platform	Not applicable	Potential recycler-side transaction fee / service fee; validate willingness-to-pay

ECONOMIC TEST

Pilot should measure “net collector benefit” = final payout – incremental travel – incremental waiting / handling cost. Adoption requires the formal route to be competitive on this net outcome, not merely on headline price.

15 • RESEARCH & VALIDATION

14. Field Research and Validation Plan

Field research is a mandatory evidence layer for this product. This PDR deliberately does not mark interviews as completed unless interview records exist. The following protocol is the minimum validation package before final submission.

RESEARCH ACTIVITY	MINIMUM SAMPLE / EVIDENCE	QUESTIONS TO ANSWER	OUTPUT
Collector interviews	≥ 2 active collectors for minimum SIH evidence; preferably 5–10 for stronger inference	Current buyers, payment behavior, average lot size, travel, price negotiation, smartphone use, language, trust barriers	Interview notes + anonymized quotes + workflow map
Shadow / observation	At least one real collection / sorting workflow if access permits	Where does delay occur? What is weighed? When is price agreed?	Observed process map + friction log
Recycler interviews	≥ 2 recyclers / receiving partners	Accepted categories, minimum lots, rate logic, pickup, registration verification, rejection reasons	Recycler intake requirements
Usability test	5+ users for prototype; include low-literacy / first-time users	Can user create and close a lot without coaching?	Task success + time-on-task + errors
Pricing backtest	Historical / pilot transactions	How close are estimates to realized prices across regions and categories?	Error metrics + confidence intervals

14.1 Evidence log template

EVIDENCE ID	SOURCE TYPE	DATE	CLAIM / INSIGHT	VALIDATION STATUS
E-001	Collector interview	TBD	Immediate settlement and nearby buyers strongly affect route choice.	To validate
E-002	Collector interview	TBD	Price transparency is a recurring information gap.	To validate
E-003	Recycler interview	TBD	Destination ranking should include accepted categories and pickup constraints.	To validate
E-004	Prototype test	TBD	Offline draft and language choice affect task completion.	To validate

16 • SUCCESS METRICS

15. Product KPIs

KPI	DEFINITION	MVP TARGET / DECISION RULE
Lot creation success	% users creating a valid lot without external help	≥ 80% in prototype usability tests before pilot
Time-to-quote	Median seconds from lot creation to reference estimate	Track by connectivity state; reduce without sacrificing comprehension

KPI	DEFINITION	MVP TARGET / DECISION RULE
Formal-routing rate	Completed lots routed to eligible formal destination / completed lots	Primary adoption outcome; establish baseline in pilot
Quote-to-final variance	reference estimate – final accepted price / final accepted price	Report median and P90; no arbitrary “accuracy” claim
Handover completion	Accepted lots that complete pickup/intake	Track by recycler and geography
Settlement latency	Time from confirmed intake to collector settlement	Target near-immediate operationally; report actual, not aspirational
Offline recovery	% offline-created lots synchronized without loss/duplication	≥ 99% in controlled pilot test
Recycler acceptance rate	Accepted lots / presented eligible lots	Use to tune matching quality

17 • IMPLEMENTATION

16. Proposed Technical Stack and Delivery Architecture

[ASSUMPTION] The stack below is an implementation recommendation for the prototype, not a mandated SIH technology selection.

AREA	RECOMMENDED PROTOTYPE CHOICE	RATIONALE
Mobile	Native Android / lightweight cross-platform app with local SQLite/Room store	Strong offline primitives and entry-level device support.
Backend	REST API with transactional service layer	Simple integration surface for mobile + recycler web.
Database	PostgreSQL	Relational integrity suits transactions, registry records and audit references.
Object storage	S3-compatible object store	Photos and evidence with controlled access and lifecycle policy.
Auth	OTP / passkey-style low-friction authentication + role-based access	Low-friction login while preserving role boundaries.
AI serving	Containerized inference endpoint or managed model API behind service abstraction	Allows model replacement without redesigning workflow.
Analytics	Event log → analytical views / dashboard	Separates operational state from metrics and experimentation.

18 • TRANSACTION DESIGN

17. Transaction State Model

STATE	ENTRY CONDITION	EXIT CONDITION	AUDIT EVENT
DRAFT	Collector starts lot	Submitted	LotCreated

STATE	ENTRY CONDITION	EXIT CONDITION	AUDIT EVENT
SUBMITTED	Lot data valid enough for matching	Quote available / clarification required	LotSubmitted
OFFERED	At least one eligible route exists	Accepted / expired	OfferShown
ACCEPTED	Collector chooses route	Pickup / handover / cancelled	OfferAccepted
IN_TRANSIT	Pickup confirmed	Received	HandoverStarted
RECEIVED	Receiver confirms intake	Settled / dispute	IntakeConfirmed
SETTLED	Payment confirmed	Closed	SettlementConfirmed
CLOSED	Settlement + evidence complete	N/A	LotClosed
DISPUTED	Any exception requiring resolution	Resolved / reversed	DisputeOpened

19 • RISKS

18. Risks, Failure Modes and Mitigations

RISK	WHY IT MATTERS	MITIGATION
Stale recycler authorization data	Wrongly presenting a non-current entity undermines trust and compliance.	Store source + verification timestamp; stale status banner; periodic sync; operator override only with evidence.
Price manipulation / lowball offers	Collector adoption can fail if marketplace is not actually competitive.	Show reference range, buyer quote source and recent transactions; log rejected / completed outcomes.
AI misclassification	A wrong category may route a lot incorrectly.	Confidence threshold + abstain state + manual correction; safety-critical flags override model suggestion.
Offline duplication	Retries can create duplicate lots / settlement events.	Idempotent event IDs, sync conflict resolution and server-side uniqueness constraints.
Cash settlement disputes	Physical cash introduces proof challenges.	Settlement confirmation by both sides + optional photo/receipt evidence + dispute workflow.
Low adoption due to habit	Existing aggregator may remain easier.	Repeat routes, minimal onboarding, cash-first flow, local language, optional assisted onboarding.
Scope creep	Trying to become a full EPR system weakens prototype delivery.	Keep statutory compliance functions out of scope; focus on collector transaction bridge.

20 • SIH ALIGNMENT

19. SIH Requirements Alignment Matrix

EXPECTED REQUIREMENT	PDR RESPONSE	DEMO EVIDENCE
Software-only solution	Mobile collector app + recycler web interface + backend	Working prototype; no dedicated hardware dependency
Offline-first support	Local lot creation, queued sync, visible sync state	Disable network and demonstrate lot creation
Vernacular / low literacy	Hindi + Marathi minimum; picture-first and voice-assisted interaction	Live task test with language switch
AI/ML where training data is sufficient	Material classification, price estimation, anomaly signals with fallbacks	Model + confidence + fallback demo
Six datasets	Material, price, recycler, transaction, traceability, collector datasets	Schema / sample data / update pipeline
AI training dataset	Lineage, labels, source quality, limitations	Dataset card + model card
Field research	Collector/recycler interviews and observations	Anonymized notes + findings
Unit economics	Collector net benefit and platform monetization hypothesis	Pilot economics sheet
Cash-first optional digital	Settlement mode selectable; digital not mandatory in UX	Complete a cash settlement scenario
Traceability	Lot ID + event chain + handover proof	Traceability timeline demo
Privacy-conscious	Minimal collector data and role-based access	Show profile fields and access controls
Recycler interface	Dedicated intake, offers, pickup, settlement and audit views	Recycler portal walk-through

21 • MVP SCOPE

20. Prototype Scope: Must / Should / Later

MUST HAVE — SIH DEMO	SHOULD HAVE — STRONGER PILOT	LATER — SCALE FEATURES
Collector onboarding	Voice guidance	Advanced route optimization
Offline lot creation	Assisted price review	Automated settlement reconciliation
Photo category suggestion	Recycler capacity feed	Fraud graph / sophisticated anomaly models
Reference price range	Cash + optional digital settlement	Multi-state tax / invoicing automation
Recycler verification metadata	Pickup scheduling	Full ERP / enterprise integrations
Traceability timeline	Basic analytics	Predictive supply planning

22 • ROADMAP

21. Delivery Roadmap

PHASE	OBJECTIVE	KEY DELIVERABLES	EXIT CRITERIA
Phase 0 • Evidence	Validate assumptions	Collector + recycler interviews; current workflow map; data sources	Top 5 friction points supported by evidence
Phase 1 • Vertical slice	Demonstrate end-to-end transaction	Collector app → lot → estimate → recycler → handover → settlement	One complete happy-path transaction works offline/online
Phase 2 • Trust layer	Increase credibility	Verification timestamps, receipts, dispute flow, safety prompts	Users understand buyer/price/settlement terms
Phase 3 • Intelligence	Add bounded AI	Classification + pricing model + anomaly rules	Models outperform baseline or abstain appropriately
Phase 4 • Pilot	Validate economic adoption	Local geography, multiple collectors/recyclers, KPI logging	Formal-routing rate and net-benefit evidence

23 • CONCLUSION

22. Conclusion

E-Waste Bridge is intentionally not positioned as “another e-waste app.” Its core product hypothesis is that formalization succeeds when the compliant route preserves the practical strengths of the existing collection economy: local reach, speed, trusted counterparties and predictable settlement. The technology then supplies what the informal transaction lacks—transparent reference value, verified destination discovery, structured handover evidence, a persistent ledger and data-driven matching.

The strongest differentiator is therefore not an isolated AI feature. It is the integration of collector-first UX, offline-first operations, verified recycler routing, cash-first settlement, traceability and a continuously improving data layer into one transaction loop. The next proof point is empirical: demonstrate with real collectors and receiving partners that the platform reduces the operational friction of choosing a formal route without making the collector economically worse off.

24 • ASSUMPTIONS & BOUNDARIES

23. Assumptions, Open Questions and Non-Claims

TYPE	STATEMENT
ASSUMPTION	Collectors in the target pilot geography have access to Android devices capable of camera capture. Validate device distribution before finalizing app footprint.
ASSUMPTION	At least a subset of eligible recyclers can provide indicative rates, accepted categories or pickup conditions digitally. Validate during partner onboarding.
ASSUMPTION	Cash settlement can remain operationally available in the target pilot; the platform does not itself need to hold funds for the MVP. Validate legal/operational model before deployment.
OPEN QUESTION	What geography and material categories should the first pilot prioritize to maximize density and transaction frequency?

TYPE	STATEMENT
OPEN QUESTION	What exact registry source/API or periodically downloaded authority data will be used to verify recycler status in the production implementation?
NON-CLAIM	The prototype does not claim that every informal collector can or should become independently authorized under the applicable regulations.
NON-CLAIM	AI estimates are not guaranteed market prices and should not be represented as such.
NON-CLAIM	The platform does not authorize hazardous processing, dismantling or metal extraction.

25 • REFERENCES

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- R9 Karo Sambhav. Collection centres / consumer drop-off information. <https://www.karosambhav.com/collection-centres>
- R10 Recykal. Marketplace overview and e-waste / traceability / compliance positioning. <https://www.recykal.com/marketplace-dc>

EVIDENCE DISCIPLINE

External sources substantiate the regulatory, health-risk and existing-ecosystem context. Product claims, field findings and performance targets remain subject to prototype and pilot validation.